Secure Elder-Care Monitoring Platform Using Video-Based Computer Vision and Schedule-Aware Activity Recognition

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Project Proposal Report

S.M.B.H.J.Jayasinghe – IT22240392

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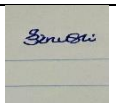
Department of Information Technology (IT)

Sri Lanka Institute of Information Technology Sri Lanka

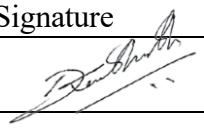
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
JAYASINGHE S M B H J	IT22240392	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
, **Dr. Dharshana Kasthurirathna**	2026/03/14	

ABSTRACT

Ensuring the safety and accountability of caregivers in elder-care environments has become increasingly important as the global elderly population continues to grow. Many families rely on CCTV monitoring to supervise caregivers and ensure the well-being of elderly individuals. However, traditional surveillance systems mainly provide visual monitoring and require manual observation, which makes it difficult to automatically verify the identity of individuals appearing in the monitored environment. Person identification technologies based on biometrics have therefore attracted significant attention in computer vision research. Facial recognition has become one of the most widely used biometric identification methods due to its convenience and high accuracy when facial information is clearly visible [1], [2]. Nevertheless, face recognition systems often experience reduced performance when faces are partially occluded, poorly illuminated, or not directly visible to the camera.

Recent studies have explored alternative biometric modalities such as gait recognition and skeleton-based identification, which analyze body structure and movement patterns to distinguish individuals [3], [4]. While these approaches can provide additional identity information, many existing systems rely on specialized sensors such as Kinect devices or depend on motion-based features that are not always available in real-world monitoring environments. As a result, current systems that rely on a single biometric modality may struggle to provide reliable person identification in practical surveillance scenarios.

To address this limitation, this research proposes a hybrid person identification system that integrates facial recognition with skeletal bone structure analysis using standard webcam-based cameras. The system will capture video streams using devices such as Logitech C270, Logitech C920, and Generic 1080p USB webcams. Face recognition techniques based on deep learning models will be combined with pose estimation methods to extract skeletal body structures from video frames [5]. Computer vision frameworks such as OpenCV will support image processing and feature extraction tasks [6].

The expected outcome of this research is a more reliable person identification system capable of identifying individuals even when facial information is partially unavailable. The proposed system can enhance security and trust in elder-care monitoring environments while reducing reliance on manual surveillance. In addition to its social benefits in improving elderly safety, the system also has potential commercial value for smart home monitoring and intelligent security applications.

Keywords: Person Identification, Face Recognition, Skeleton-Based Identification, Computer Vision, Elder-Care Monitoring.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence
API – Application Programming Interface
CNN – Convolutional Neural Network
CV – Computer Vision
CVPR – Conference on Computer Vision and Pattern Recognition
DL – Deep Learning
F1-score – Harmonic Mean of Precision and Recall
FPS – Frames Per Second
GPU – Graphics Processing Unit
HCI – Human–Computer Interaction
IoT – Internet of Things
ML – Machine Learning
OpenCV – Open Source Computer Vision Library
RGB – Red Green Blue (Color Image Format)
REST – Representational State Transfer
SDG – Sustainable Development Goal
UI – User Interface
USB – Universal Serial Bus
WBS – Work Breakdown Structure
YOLO – You Only Look Once (Object Detection Algorithm)

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1. INTRODUCTION

1.1 Background and Context

The global population is rapidly aging, creating increasing demand for reliable monitoring and security systems in elder-care environments. Families and healthcare providers increasingly depend on caregivers to support elderly individuals with daily activities such as medication management, mobility assistance, and personal care. However, ensuring the safety, authenticity, and accountability of caregivers remains a significant challenge in home-care settings. In many cases, families rely on CCTV systems to monitor caregivers remotely. While CCTV cameras provide visual monitoring, they still require manual observation and cannot automatically verify the identity of individuals appearing in the surveillance footage.

Automated person identification technologies have therefore become an important research area in computer vision and artificial intelligence. Person identification refers to the process of recognizing or verifying the identity of an individual using biometric characteristics. Biometric systems commonly use physiological or behavioral traits such as facial features, fingerprints, iris patterns, voice, or gait for identifying individuals [7]. Among these, facial recognition has emerged as one of the most widely adopted biometric identification methods due to its convenience and ability to operate using standard camera systems.

Recent advances in deep learning have significantly improved the performance of facial recognition systems. Modern approaches such as FaceNet and ArcFace use deep convolutional neural networks to extract highly discriminative facial embeddings that allow accurate identification of individuals even in complex environments [1], [2]. Deep neural network architectures, particularly convolutional neural networks (CNNs), have become fundamental to many computer vision tasks, including image classification and recognition [8]. These advancements have enabled face recognition systems to achieve high accuracy and scalability in applications such as surveillance, authentication, and smart security systems.

Despite these improvements, facial recognition systems still face several practical limitations in real-world surveillance scenarios. Factors such as poor lighting conditions, camera angle variations, facial occlusions, and motion blur can significantly reduce recognition accuracy. In many monitoring environments, including elder-care homes, individuals may not always face the camera directly, making facial recognition unreliable. Therefore, relying solely on

facial features for person identification may not provide sufficient robustness in practical applications.

To overcome these challenges, researchers have explored alternative biometric modalities such as gait and skeletal body structures. Human gait and skeletal characteristics represent unique physical patterns associated with body structure, limb proportions, and walking styles. These features are difficult to disguise and remain relatively stable over time, making them useful for identifying individuals even when facial information is unavailable. Research has shown that individuals can be reliably identified by analyzing gait patterns captured from video sequences [4].

Skeleton-based person identification has recently gained attention as an emerging research direction in computer vision. This approach extracts skeletal joint positions from video frames and analyzes body geometry and motion patterns to distinguish individuals. Skeleton-based identification methods have demonstrated promising results in recognizing individuals using body structure features and movement dynamics [3]. Furthermore, recent research surveys highlight the growing importance of skeleton-based person re-identification techniques for applications such as surveillance, robotics, and smart environments [9].

Many skeleton-based identification approaches rely on analyzing body joint movements captured through specialized sensors such as depth cameras or Kinect devices. These methods use skeletal joint coordinates to derive structural and dynamic features that can uniquely characterize individuals [3]. Studies on gait recognition have shown that skeletal features such as stride length, joint angles, and limb proportions can effectively distinguish individuals even when facial data is not available [10]. These findings demonstrate that skeletal body structures provide a complementary biometric modality to facial recognition.

Modern computer vision systems also incorporate pose estimation techniques to extract skeletal joint information from standard RGB cameras. Techniques such as real-time multi-person pose estimation enable the detection of body joints and skeletal structures from video frames without requiring specialized depth sensors [5]. Additionally, object detection frameworks such as YOLO and advanced deep learning architectures like ResNet have further improved the efficiency and accuracy of real-time computer vision systems [11], [12]. These technologies make it possible to implement intelligent surveillance systems using standard camera hardware.

Computer vision libraries such as OpenCV provide powerful tools for image processing, object detection, and feature extraction, enabling developers to implement complex visual recognition systems efficiently [6]. Furthermore, open-source frameworks such as OpenFace allow the extraction of detailed facial behavior and facial landmark features for facial recognition tasks [13]. Advances in lightweight neural network architectures such as MobileNet also enable efficient deployment of deep learning models on resource-constrained devices and edge computing environments [14]. Object detection improvements such as single-shot refinement neural networks further enhance the performance of real-time detection systems used in surveillance applications [15].

Although significant research has been conducted in facial recognition and skeleton-based identification independently, many existing systems rely on only one biometric modality. Systems that depend solely on facial recognition may fail when the face is partially visible or occluded, while systems relying only on gait or skeletal data may struggle in crowded or stationary scenarios. Therefore, integrating multiple biometric modalities can significantly improve identification accuracy and robustness.

To address these challenges, this research proposes a **hybrid person identification approach that combines facial features and skeletal bone structure analysis using standard webcam-based surveillance systems**. By integrating facial recognition with skeletal pose analysis, the system can identify individuals even when facial features are partially hidden or unclear. The proposed system will capture video streams using commonly available cameras such as **Logitech C270 Webcam, Logitech C920 Webcam, and Generic 1080p USB Webcam**. Facial recognition algorithms will first attempt to verify the individual's identity. If facial information is insufficient, the system will analyze skeletal body structures and pose information to support identification.

The integration of facial and skeletal features provides a more reliable and robust person identification framework for surveillance environments. This approach is particularly valuable in elder-care monitoring systems, where continuous observation and identity verification are necessary to ensure the safety and well-being of elderly individuals. By leveraging modern computer vision techniques and widely available camera hardware, the proposed system aims to improve monitoring efficiency while reducing reliance on manual supervision.

Furthermore, this research aligns with the **Artificial Intelligence and Computer Vision research cluster**, as it focuses on biometric recognition, machine learning, and intelligent surveillance systems. The project also contributes to **Sustainable Development Goal (SDG) 3: Good Health and Well-being**, which aims to ensure healthy lives and promote well-being for people of all ages. By improving monitoring technologies in elder-care environments, the proposed solution supports safer living conditions and enhances trust between families, caregivers, and healthcare providers.

Overall, the proposed system aims to develop an intelligent and scalable person identification framework that integrates facial recognition and skeletal body structure analysis to provide reliable identity verification in real-time monitoring environments.

1.2 Literature Review

Person identification has become an important research area in computer vision, especially for surveillance and healthcare monitoring systems. Modern identification systems aim to automatically recognize individuals using biometric information extracted from images or video. In environments such as elder-care monitoring, these systems can help verify the

identity of caregivers and detect unknown individuals in the monitored area. An automated identification system generally involves several steps including capturing visual data, extracting meaningful features, and comparing those features with stored identity information.

Most modern person identification systems rely on artificial intelligence and deep learning techniques. Cameras capture images or video of individuals, and computer vision algorithms analyze this data to extract biometric characteristics such as facial features or body structure. Machine learning models are then used to match these features with known identities. Many systems combine several technologies such as face recognition, pose estimation, and object detection to improve reliability in real-world environments.

Face recognition is one of the most widely used biometric identification methods. Deep learning models have greatly improved the accuracy of face recognition systems. FaceNet introduced a method that converts facial images into numerical feature representations that can distinguish individuals effectively [2]. ArcFace later improved recognition performance by introducing an enhanced training strategy that increases the discrimination between different faces [1]. These approaches are built on convolutional neural networks, which became widely used after deep learning models achieved major success in image classification tasks [8]. Because of these developments, face recognition systems are now used in many security and monitoring applications.

Researchers have also studied other biometric characteristics to support person identification when facial information is not clearly visible. One important approach is skeleton-based identification, which analyzes the structure and movement of the human body. Instead of focusing only on facial appearance, this method examines body joints, limb lengths, and walking patterns to identify individuals. Studies show that skeletal body structures and movement patterns can contain unique features that help distinguish different people [9].

Gait recognition is one of the earliest techniques used for body-based identification. Kale et al. demonstrated that people can be recognized by analyzing their walking style and movement patterns captured from video sequences [4]. Later studies explored the use of depth sensors and skeletal joint data to improve identification accuracy. Preis et al. developed a gait recognition method using Kinect sensors to track body joints and analyze walking patterns [10]. Similarly, Sinha et al. proposed a skeleton-based identification approach that extracts both structural and motion features from body joints to recognize individuals [3].

Recent developments in computer vision have made it possible to detect skeletal structures directly from regular camera images. Pose estimation algorithms can detect body joints and create a skeleton representation of a person in a video frame. Cao et al. introduced a real-time pose estimation method that can detect multiple human skeletons from images and video [5]. This allows skeleton-based identification systems to be implemented using normal cameras instead of specialized depth sensors.

Object detection techniques also play a role in modern identification systems. Models such as YOLO enable real-time detection of people and objects in video streams [12]. Deep learning architectures such as ResNet have improved recognition accuracy by enabling deeper neural networks that learn more complex visual features [11]. Other detection methods, such as single-shot refinement neural networks, have further improved object detection accuracy in challenging environments [15].

Several software tools support the development of computer vision identification systems. OpenCV is widely used for image processing and computer vision applications [6]. OpenFace provides open-source tools for facial landmark detection and facial behavior analysis, which are useful for face recognition systems [13]. Lightweight neural network models such as MobileNet also enable computer vision systems to run efficiently on devices with limited computing resources [14].

Although significant progress has been made in face recognition and skeleton-based identification, current systems still have limitations. Face recognition performance can decrease when faces are partially hidden, poorly illuminated, or viewed from unusual angles. Skeleton-based systems also depend on accurate detection of body joints and may perform poorly when parts of the body are occluded or when a person is not moving.

Another limitation is that many systems rely on only one biometric feature. Systems that depend only on face recognition may fail when facial information is unclear, while systems based only on skeletal features may struggle when individuals are standing still or when body movement is limited. Because of these challenges, researchers are increasingly exploring methods that combine multiple biometric features to improve identification accuracy and reliability.

1.3 Research Gap

Existing research has made significant progress in person identification using biometric technologies such as facial recognition, gait analysis, and skeleton-based identification. Deep learning models such as FaceNet and ArcFace have demonstrated very high accuracy in recognizing individuals from facial images by learning discriminative facial features [1], [2]. These systems rely on convolutional neural networks that extract complex visual patterns from large image datasets [8]. As a result, face recognition has become one of the most widely adopted biometric identification methods in surveillance systems, authentication platforms, and security applications.

However, face recognition systems are highly dependent on the visibility and quality of facial information. In real-world monitoring environments, especially in surveillance or elder-care systems, faces may not always be clearly visible due to lighting variations, occlusions, head movements, or camera angles. When the face is partially hidden or turned away from the camera, the accuracy of face recognition systems can significantly decrease. This limitation

indicates that relying solely on facial features may not provide reliable person identification in practical monitoring scenarios.

To address these limitations, researchers have explored body-based identification techniques such as gait recognition and skeleton-based person identification. Gait recognition methods analyze walking patterns to identify individuals based on their movement style [4]. Similarly, skeleton-based approaches use body joint positions and structural body features to distinguish individuals [3]. Some studies have used depth sensors such as Kinect to capture skeletal joint data and extract both structural and dynamic features for person identification [10]. These approaches demonstrate that body structure and motion patterns contain useful identity-related information.

Although skeleton-based identification methods provide advantages in situations where facial information is unavailable, they also have limitations. Many existing studies rely on specialized sensors such as Kinect devices to obtain accurate skeletal joint data [3], [10]. These sensors are not always practical for real-world deployment in home environments or standard surveillance systems. In addition, skeleton-based identification methods often depend on motion-based features such as gait patterns, which means they may perform poorly when individuals are standing still or when body movement is limited.

Recent advances in pose estimation algorithms allow skeletal joint positions to be extracted from regular camera images without specialized sensors [5]. However, most research studies still treat facial recognition and skeleton-based identification as separate approaches rather than integrating them into a unified system. Systems based solely on face recognition may fail when facial features are unclear, while skeleton-based methods may struggle when body motion is limited or when pose estimation is inaccurate.

This comparison highlights an important research gap. While both facial recognition and skeleton-based identification techniques have been widely studied, there is limited research on integrating these two biometric modalities to improve person identification reliability in real-world surveillance environments. Existing systems typically rely on a single type of biometric feature, which reduces robustness when environmental conditions change.

Therefore, there is a need for a hybrid person identification approach that combines multiple biometric features to overcome the limitations of single-modality systems. Integrating facial features with skeletal body structure analysis can improve identification reliability by allowing the system to use alternative biometric cues when one feature is unavailable. For example, when facial features are partially occluded, skeletal body structure can assist in identifying the individual. Conversely, when body pose information is insufficient, facial recognition can provide the necessary identification evidence.

The proposed research addresses this gap by developing a **person identification system that integrates facial recognition and skeletal bone structure analysis using standard webcam-based surveillance cameras**. Unlike many existing skeleton-based systems that depend on depth sensors, the proposed approach uses pose estimation techniques to extract

skeletal information from regular RGB camera footage. This allows the system to be deployed using commonly available devices such as **Logitech C270 Webcam, Logitech C920 Webcam, and Generic 1080p USB Webcam**.

By combining facial features and skeletal body structure information within a single identification framework, the proposed system aims to improve identification reliability in real-world monitoring environments. This multimodal approach represents a creative extension of existing research by integrating two complementary biometric modalities within a practical surveillance system designed for elder-care monitoring applications.

2. OBJECTIVE

2.1 Main Objective

To design and evaluate a hybrid person identification system that integrates facial recognition and skeletal bone structure analysis using webcam-based surveillance cameras to improve identification accuracy in elder-care monitoring environments.

2.2 Specific Objectives

1. To develop a facial recognition module capable of detecting and identifying individuals from video frames.
This objective focuses on implementing a facial recognition system that can detect human faces in real-time video streams captured by webcams. The module will extract facial features from detected faces and compare them with stored identity data to verify individuals. The effectiveness of this module will be evaluated based on recognition accuracy and the system's ability to identify individuals under different lighting conditions and camera angles.

2. To implement a skeletal pose detection module that extracts human skeletal joint structures from-RGB-video-streams**.**
This objective aims to develop a pose estimation component that detects body joints and constructs a skeletal representation of individuals captured in the video. By analyzing skeletal joint positions and body structure, the system will generate additional biometric features that can be used for person identification when facial information is not clearly visible.

3. To design and integrate a multimodal identification framework that combines facial features and-skeletal-body-structure-information**.**
This objective focuses on integrating the facial recognition module and skeletal pose detection module into a single identification framework. The system will analyze both facial and skeletal features and determine the identity of individuals by combining the information from both sources, thereby improving identification reliability when one modality is insufficient.

4. To evaluate the performance of the proposed hybrid identification system using quantitative performance-metrics**.**

This objective involves testing the system under different conditions to measure its effectiveness. Performance will be evaluated using metrics such as identification accuracy, precision, recall, and processing time. These evaluations will help determine how reliably the system can identify individuals in practical monitoring environments.

5. To compare the performance of the proposed hybrid system with single-modality identification approaches.
This objective aims to analyze whether combining facial recognition and skeletal body structure analysis improves identification accuracy compared to systems that rely only on facial recognition or skeletal features. The comparison will demonstrate the advantages of the hybrid approach in improving robustness and reliability in real-world monitoring scenarios.

3. METHODOLOGY

3.1 Research Approach

This research adopts a **System Development and Experimental Evaluation approach** to design and validate a person identification system for elder-care monitoring environments. The goal is to develop a practical solution that can automatically identify individuals using visual information captured from surveillance cameras.

The study involves designing a hybrid identification framework that integrates facial recognition and skeletal body structure analysis. After implementing the system, controlled experiments will be conducted to evaluate its ability to correctly identify individuals in real-world scenarios. The proposed system will also be compared with single-modality identification approaches to determine whether combining multiple biometric features improves identification reliability.

3.2 Technical Methods and Algorithms

The proposed system uses several computer vision and deep learning algorithms to perform multimodal person identification. Each algorithm plays a specific role in the processing pipeline.

Person Detection using YOLO

The system first detects individuals in video frames using the YOLO (You Only Look Once) object detection algorithm. YOLO is a real-time detection model that divides an image into grid cells and predicts bounding boxes and class probabilities for objects within each grid. In this research, the YOLO model will detect human subjects in the video stream and generate bounding boxes representing detected persons. This step ensures that subsequent processing stages focus only on regions containing people, improving computational efficiency and reducing false detections.

Face Detection and Recognition

After a person is detected, the system performs face detection to identify facial regions within the detected bounding box. Detected faces are processed using deep learning–based face recognition models such as FaceNet or ArcFace.

FaceNet converts facial images into numerical feature vectors known as **facial embeddings**. These embeddings represent distinctive facial characteristics in a high-dimensional feature space. During identification, the system compares the extracted facial embedding with stored embeddings in the database using similarity metrics such as cosine similarity or Euclidean distance.

ArcFace improves recognition accuracy by introducing an additive angular margin loss function during training. This method increases the separation between different identities in the embedding space, enabling more reliable face recognition even in challenging conditions.

Skeletal Pose Estimation

To provide an alternative biometric modality, the system also performs skeletal pose

estimation using pose estimation frameworks such as OpenPose or MediaPipe Pose. These algorithms detect key body joints such as shoulders, elbows, hips, and knees from RGB images.

The pose estimation model generates a skeletal representation consisting of joint coordinates. From these coordinates, structural features such as limb lengths, joint angles, and body proportions can be extracted. These skeletal features provide identity-related information that can support identification when facial features are not clearly visible.

Skeletal Feature Extraction

Once skeletal joints are detected, the system computes several structural features including:

- Joint distance ratios
- Limb length proportions
- Relative joint angles
- Body posture patterns

These features form a skeletal feature vector representing the physical structure of the individual. Because body proportions tend to remain stable over time, they can serve as a supplementary biometric characteristic.

Multimodal Feature Fusion

The final identification decision is made using a multimodal feature fusion approach that combines facial embeddings and skeletal features. Feature fusion improves system robustness by allowing the system to rely on alternative biometric information when one modality becomes unreliable.

Two fusion strategies may be considered:

1. **Score-level fusion** – combining similarity scores from face recognition and skeletal analysis.
2. **Decision-level fusion** – selecting the most confident prediction from the available modalities.

By integrating facial and skeletal features, the proposed system aims to achieve more reliable identification compared to single-modality approaches.

3.3 Tools and Platforms with Justification

The proposed system will be implemented using several software tools and development platforms selected for their suitability in computer vision applications.

Python will be used as the primary programming language because it provides extensive support for machine learning and computer vision development.

The OpenCV library will be used for image processing, video frame extraction, and real-time camera integration [6]. OpenCV provides efficient functions for detecting faces, handling video streams, and processing image data.

Deep learning models will be implemented using frameworks such as TensorFlow or PyTorch, which support training and deploying neural networks efficiently. Lightweight architectures such as MobileNet may also be used to ensure the system can operate efficiently on systems with limited computational resources [14].

Facial feature extraction may be supported by the OpenFace framework, which provides tools for detecting facial landmarks and extracting facial features [13].

For data storage, a database system such as MySQL or a NoSQL database may be used to store identity information, facial embeddings, and system logs.

The system will use standard webcam devices including Logitech C270, Logitech C920, and Generic 1080p USB webcams to capture video streams. Using widely available hardware ensures that the proposed solution can be deployed in real-world monitoring environments without requiring specialized sensors.

3.4 Data Collection and Ethical Considerations

The dataset for this research will be collected using video recordings captured from the selected webcam devices in controlled indoor environments. A group of volunteer participants will be recorded while performing normal movements such as walking, standing, and interacting with the environment. These recordings will provide facial images and skeletal body structures that will be used for training and evaluating the identification system.

Ethical considerations will be strictly followed during the data collection process. Participation will be voluntary, and individuals will be informed about the purpose of the research before recording begins. Participants will provide consent before their data is collected.

Personal data will be handled responsibly, and the collected recordings will be used only for research purposes. The dataset will be stored securely, and any personal information will be protected to maintain participant privacy.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed system will be evaluated through experimental testing using recorded video samples. The experiments will measure how accurately the system can identify individuals under different environmental conditions.

Several performance metrics will be used to evaluate the system. Accuracy will measure the percentage of correctly identified individuals. Precision and recall will be used to analyze how effectively the system detects and identifies individuals. The F1-score will be calculated to provide a balanced measure of identification performance.

In addition, system response time will be measured to evaluate whether the system can operate in real-time monitoring environments.

To demonstrate the effectiveness of the proposed hybrid approach, the performance of the multimodal system will be compared with systems that use only facial recognition or only skeletal features.

Limitations and Challenges

Although the proposed hybrid identification system offers several advantages, certain limitations and challenges may affect system performance.

Lighting and Environmental Conditions

Computer vision algorithms may experience reduced accuracy in environments with poor lighting, shadows, or strong illumination variations. These factors can affect both facial recognition and pose estimation accuracy.

Pose Estimation Errors

Pose estimation models may produce incorrect joint detections when body parts are occluded or when multiple individuals appear in the camera frame. Incorrect skeletal joint detection can reduce the reliability of skeletal feature analysis.

Limited Dataset Size

The system relies on a dataset collected from a limited number of participants. A small dataset may restrict the generalization ability of the trained models and reduce performance when applied to larger populations.

Real-Time Processing Constraints

Processing video frames using deep learning models requires computational resources. Systems running on limited hardware may experience delays that affect real-time monitoring performance.

Privacy and Ethical Concerns

Because the system processes biometric data such as facial features and skeletal structures, privacy concerns may arise. Strict data protection mechanisms and informed consent procedures are necessary to ensure ethical use of the system.

Despite these limitations, the multimodal design of the proposed system helps mitigate several challenges by combining complementary biometric features.

3.6 Evaluation Plan

To assess the effectiveness of the proposed hybrid person identification system, a structured evaluation plan will be implemented. The evaluation will measure the system's identification accuracy, robustness, and real-time performance in simulated elder-care monitoring environments.

Dataset Preparation

Video data will be collected from volunteer participants using webcam cameras such as Logitech C270, Logitech C920, and Generic 1080p webcams. Participants will be recorded while performing common activities such as walking, standing, and interacting with the environment. The dataset will include variations in lighting conditions, camera angles, body poses and movements, and situations where facial features are partially occluded.

The collected dataset will be divided into three subsets. The training dataset, representing approximately 70% of the collected data, will be used to train the facial recognition and skeletal feature models. A validation dataset containing around 15% of the data will be used to tune model parameters and optimize performance. The remaining 15% of the dataset will be used as the testing dataset for the final performance evaluation of the system.

Experimental Scenarios

Several experimental scenarios will be used to evaluate system performance under realistic monitoring conditions. In the normal identification scenario, individuals appear clearly in the camera view with visible facial features, allowing the system to perform standard identification. In the face occlusion scenario, the individual's face may be partially hidden or turned away from the camera, testing the system's ability to rely on skeletal features when facial information is limited. In the body pose variation scenario, individuals appear in different body poses such as walking, standing, or bending, allowing the system to evaluate identification performance under varying skeletal structures. Another scenario includes situations where multiple individuals appear in the camera frame simultaneously, enabling the evaluation of detection and identification accuracy in multi-person environments. These scenarios simulate realistic elder-care monitoring situations where environmental conditions and human behavior may vary.

Baseline Comparison

To demonstrate the advantages of the proposed hybrid identification approach, the system will be compared with two baseline models. One baseline system will rely solely on facial recognition for person identification, while the second baseline system will rely only on skeletal feature–based identification. By comparing these systems with the proposed hybrid

model, the research will determine whether combining facial and skeletal features improves overall identification accuracy and robustness.

Performance Metrics

The system will be evaluated using several quantitative performance metrics. Accuracy will measure the percentage of correctly identified individuals. Precision will evaluate how many of the predicted identities are correct, while recall will measure how effectively the system identifies all relevant individuals in the dataset. The F1-score will be calculated to provide a balanced measure that combines precision and recall. In addition, system processing time will be measured in frames per second (FPS) to determine whether the system can operate effectively in real-time monitoring environments.

Evaluation Procedure

During the evaluation process, video frames captured from webcam cameras will be processed by the system. The object detection module will first identify the presence of individuals within each frame. Facial features will then be extracted using the face recognition model, while skeletal joint features will be extracted using pose estimation techniques. The system will then combine facial embeddings and skeletal features through the multimodal feature fusion mechanism to determine the identity of the individual. The predicted identity will be compared with the ground truth identity stored in the dataset, and performance metrics will be calculated for each experimental scenario.

The results obtained from these experiments will provide a comprehensive assessment of the effectiveness, robustness, and real-time performance of the proposed hybrid person identification system.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is designed as a **multimodal person identification framework** that combines facial recognition and skeletal body structure analysis to identify individuals in an elder-care monitoring environment. The system operates using standard webcam devices such as Logitech C270, Logitech C920, and Generic 1080p USB webcams to capture real-time video streams. The captured video data is processed using computer vision algorithms to detect individuals and extract biometric features.

The architecture consists of several components including video acquisition, person detection, facial recognition, skeletal pose estimation, feature integration, identity verification, and data storage. These components work together to analyze visual information and determine the identity of individuals present in the monitoring environment. The integration of multiple biometric features improves system reliability when one feature is

unavailable or unclear.

4.2 System Components

The proposed system is composed of several functional modules that work together to perform person identification.

The **Video Acquisition Module** is responsible for capturing video streams from the connected webcam devices. This module continuously records video frames and forwards them to the processing pipeline for analysis.

The **Person Detection Module** analyzes the captured frames to detect the presence of individuals. Object detection algorithms identify human subjects in the scene and isolate the relevant regions of interest for further processing.

The **Facial Recognition Module** processes the detected face region to extract facial features. Deep learning-based face recognition models generate facial embeddings that represent unique facial characteristics of individuals. These embeddings are compared with stored identity data to verify the person's identity.

The **Skeletal Pose Estimation Module** extracts body joint positions from the video frames. The module constructs a skeletal representation of the detected person and analyzes body structure features that can assist in identification when facial information is insufficient.

The **Feature Integration Module** combines facial features and skeletal body structure information. This multimodal fusion step improves identification reliability by allowing the system to use multiple biometric cues.

The **Identity Verification Module** compares the extracted features with stored identity profiles in the system database. Based on similarity scores, the system determines the most likely identity of the individual.

The **Database and Storage Module** stores identity profiles, facial embeddings, skeletal feature data, and system logs. This module also maintains records of detected individuals for monitoring and auditing purposes.

4.3 System Diagram

The following conceptual diagram represents the high-level architecture of the proposed system.

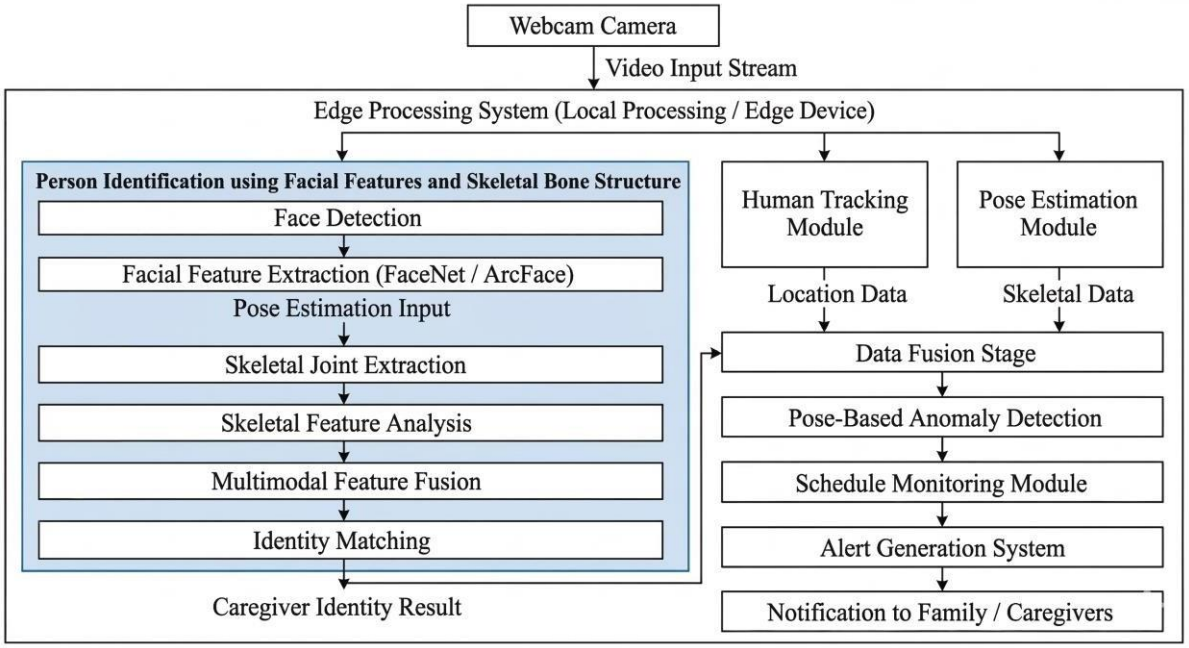


Figure 1:System Diagram

4.4 Component Interactions

As shown in Figure 4.3, the proposed system processes video streams captured from webcam cameras through several modules to perform caregiver identification and monitoring. The process starts with the Webcam Camera, which captures continuous video footage from the monitored environment. This video stream is sent to the Video Input Stream, where the frames are prepared for processing in the Edge Processing System.

Within the edge processing layer, several modules operate together. The main research component is the Person Identification using Facial Features and Skeletal Bone Structure module, which analyzes both facial and skeletal biometric information. The system first performs face detection to locate faces in the video frames. The detected faces are then processed by the Facial Feature Extraction stage, where deep learning models such as FaceNet or ArcFace generate facial embeddings representing the individual.

At the same time, the system processes body posture information through pose estimation input, allowing skeletal joint positions to be extracted. The Skeletal Joint Extraction module detects key body joints, and the Skeletal Feature Analysis module analyzes structural body features. These facial and skeletal features are combined in the Multimodal Feature Fusion stage, allowing the system to use both types of biometric information. The Identity Matching module then compares these features with stored identity profiles and produces the Caregiver Identity Result.

Alongside this component, the Human Tracking Module monitors the movement and location of individuals and produces Location Data, while the Pose Estimation Module generates Skeletal Data related to body posture.

These outputs are sent to the Data Fusion Stage, where identity information, location data, and skeletal data are combined. The combined data is analyzed by the Pose-Based Anomaly Detection module to detect unusual or unsafe activities. The results are then checked by the Schedule Monitoring Module to see whether caregiver activities match predefined schedules. If abnormal behavior or schedule deviations are detected, the Alert Generation System sends notifications to family members or caregivers through the notification module.

4.5 Data Flow Explanation

The data flow of the proposed system begins with video captured from the surveillance cameras, as shown in Figure 4.3. The webcams continuously record video streams, which are sent to the Video Input Stream. These video frames are then processed locally within the Edge Processing System, enabling real-time analysis.

The system first performs facial analysis within the Person Identification module. Facial features are extracted and converted into numerical representations using deep learning models. At the same time, pose estimation processes the video frames to extract skeletal joint coordinates.

The skeletal joint data is analyzed to produce body structure features representing the person's posture and body proportions. These skeletal features are then combined with facial embeddings in the Multimodal Feature Fusion stage. The combined features are compared with stored identity profiles in the Identity Matching module, which generates the Caregiver Identity Result.

Meanwhile, the Human Tracking Module produces location information, and the Pose Estimation Module generates skeletal posture data. These outputs are combined in the Data Fusion Stage, where identity, location, and skeletal information are integrated.

The combined data is then analyzed by the Pose-Based Anomaly Detection module, which detects unusual movements or behavior patterns. The results are checked by the Schedule Monitoring Module to determine whether caregiver actions follow predefined schedules. If a deviation is detected, the Alert Generation System triggers notifications that are sent to family members or caregivers.

4.6 Scalability and Security Considerations

The proposed system is designed to be scalable so it can be used in larger monitoring environments such as elder-care facilities or nursing homes. Additional webcams can be added to monitor multiple locations at the same time. Since the system processes data within the **Edge Processing System**, it can handle multiple video streams while maintaining real-

time performance. The modular architecture also allows new identification methods or analytical modules to be added in the future without major changes to the system.

Security and privacy are also important because the system handles biometric data such as facial features and skeletal information. The system will use secure access control to protect stored identity profiles and monitoring records. Facial embeddings, skeletal features, and system logs will only be accessible to authorized users.

Ethical guidelines will also be followed during data collection and system use. Participants will be informed about the purpose of the monitoring system, and consent will be obtained before collecting video data. These measures help protect personal information while ensuring the system operates responsibly.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed elder-care monitoring system involves several stakeholders who interact with or benefit from the system in different ways. The primary stakeholders are elderly individuals, whose safety and well-being are supported through continuous monitoring of caregiver activities. Family members of the elderly are also key stakeholders, as they receive notifications and alerts when abnormal activities or unexpected events occur. Caregivers interact with the system indirectly, as their identities are verified and their activities are monitored according to predefined schedules. In addition, healthcare providers or facility administrators may use the system to monitor caregiving activities and ensure that proper care is provided to elderly individuals. Finally, system administrators or technical personnel are responsible for managing the system, maintaining identity records, and ensuring that the monitoring platform operates reliably and securely.

5.2 Functional Requirements

Video-Capture
The system shall capture real-time video streams from connected webcam cameras such as Logitech C270, Logitech C920, or other compatible USB webcams. These cameras will continuously record the monitoring environment and provide video frames that will be used as input for further processing by the system.

Person-Detection
The system shall analyze incoming video frames to detect the presence of individuals in the

monitored environment. Computer vision algorithms will identify human figures in the video stream and isolate the relevant regions of interest so that further analysis can be performed only on detected individuals.

Facial-Recognition

The system shall detect faces within the captured video frames and extract facial features using deep learning-based face recognition models. The extracted facial embeddings will be compared with stored identity profiles in order to verify the identity of caregivers present in the environment.

Skeletal-Structure-Extraction

The system shall extract skeletal body structures from video frames using pose estimation techniques. Body joint positions such as shoulders, elbows, knees, and other key points will be detected to generate a skeletal representation of the person.

Multimodal-Person-Identification

The system shall combine facial features and skeletal body structure information to perform reliable person identification. By integrating both biometric modalities, the system can identify individuals even when facial information is partially unclear or occluded.

Huma-Tracking

The system shall track the movement and location of detected individuals within the monitored environment. This functionality allows the system to understand the spatial position of caregivers and observe movement patterns over time.

Activity-Monitoring

The system shall analyze body posture and skeletal movement patterns to identify abnormal or unusual activities. By evaluating pose data and movement behavior, the system can detect situations that may indicate unsafe conditions for the elderly.

Schedule-Monitoring

The system shall compare detected caregiver activities with predefined caregiving schedules. This allows the system to determine whether expected caregiving tasks are performed within the scheduled time periods.

Alert-Generation

The system shall generate alerts when abnormal activities, unexpected behavior patterns, or schedule deviations are detected. These alerts help notify responsible parties about potential issues that require attention.

NotificationSystem

The system shall send notifications to family members or caregivers when alerts are triggered. Notifications may be delivered through digital communication channels such as mobile applications, messaging services, or monitoring dashboards.

5.3 Non-Functional Requirements

Security

The system must ensure that sensitive data such as facial features, skeletal information, and identity records are protected from unauthorized access. Access to stored data and system functions should be restricted to authorized users through secure authentication and access control mechanisms.

Performance

The system should process video frames and perform person identification with minimal delay in order to support near real-time monitoring. Efficient algorithms and optimized processing methods should be used to ensure that detection, identification, and anomaly analysis are completed quickly.

Reliability

The system should operate continuously and maintain stable performance during long monitoring periods. It should minimize system failures or interruptions and ensure that monitoring tasks are performed consistently.

Usability

The system interface should be easy to understand and operate for users such as family members and system administrators. Users should be able to view alerts, monitoring results, and system information without requiring advanced technical knowledge.

Scalability

The system should support the addition of more cameras or monitoring locations if the monitoring environment expands. The system architecture should allow new devices or modules to be integrated without major modifications.

Privacy

The system should ensure that video recordings and biometric data are handled responsibly. Collected data should only be used for monitoring purposes, and proper consent should be obtained from individuals whose data is recorded.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed system targets organizations and individuals who require reliable monitoring solutions for elderly care environments. The primary target market includes elder-care facilities, nursing homes, home-care service providers, and families caring for elderly individuals at home. These users often require systems that can ensure caregiver accountability, detect abnormal situations, and provide real-time monitoring of caregiving activities.

The typical customer persona includes family members responsible for elderly relatives, care facility administrators, and healthcare service providers who want to improve monitoring and safety within caregiving environments. These users are generally concerned about the safety of elderly individuals and want a reliable system that can automatically identify caregivers, monitor activities, and generate alerts when unusual events occur.

6.2 Value Proposition

The proposed system provides a smart monitoring solution that combines facial recognition and skeletal body structure analysis to identify caregivers and detect abnormal activities in elder-care environments. Unlike traditional CCTV monitoring systems that require manual observation, the proposed system performs automated person identification and behavior monitoring.

By integrating multiple biometric features, the system improves identification reliability even when facial information is partially hidden. This helps families and healthcare providers ensure that authorized caregivers are present and that caregiving activities follow expected schedules. The system therefore provides increased safety, transparency, and trust in caregiving services.

6.3 Revenue Model

The system can generate revenue through a software-based monitoring platform combined with hardware installation services. One possible model is a subscription-based service, where users pay a monthly or yearly fee to access the monitoring system, receive alerts, and maintain caregiver identity records.

Another revenue option is a one-time system installation fee, where customers purchase the monitoring system together with camera setup and system configuration. Additional revenue may also be generated through maintenance services, system upgrades, and extended support packages.

6.4 Cost Estimation

The main costs involved in developing and deploying the system include hardware costs, software development costs, and maintenance expenses. Hardware costs include webcam cameras, edge processing devices, and network infrastructure required to support video monitoring. Software development costs include the development of computer vision algorithms, system integration, and testing.

Operational costs may include system maintenance, software updates, data storage, and technical support services. By using commonly available webcam cameras and open-source software frameworks, the overall cost of the system can be kept relatively affordable.

6.5 Pricing Strategy

The pricing strategy should balance affordability and sustainability. A tiered pricing model can be used depending on the size of the monitoring environment and the number of cameras installed. For example, small home-care environments may require only a basic package with limited cameras, while larger elder-care facilities may require advanced monitoring packages with additional cameras and monitoring features.

Subscription-based pricing can also provide flexibility for customers while ensuring continuous revenue for system maintenance and upgrades.

6.6 Competitive Advantage

The proposed system offers several advantages compared to traditional monitoring systems. Most existing monitoring solutions rely only on CCTV cameras and manual observation, which can be inefficient and unreliable. The proposed system introduces automated person identification using both facial recognition and skeletal body structure analysis, which improves identification accuracy in real-world environments.

Another advantage is the use of standard webcam cameras instead of specialized hardware, which reduces system cost and makes the solution easier to deploy. In addition, the integration of caregiver identity verification, human tracking, anomaly detection, and schedule monitoring provides a more comprehensive monitoring system compared to many existing solutions.

6.7 Intellectual Property Considerations

The intellectual property of the proposed system mainly lies in the software architecture, identification algorithms, and integration of facial and skeletal biometric features for elder-care monitoring. If the system demonstrates unique technical contributions, the developers may consider protecting the solution through software copyrights, patents, or proprietary system designs.

Protecting intellectual property can help prevent unauthorized use of the developed technology and support commercialization efforts. In addition, licensing agreements may be used if the system is distributed to healthcare organizations or commercial service providers.

7. BUDGET AND JUSTIFICATION

The development and implementation of the proposed elder-care monitoring system require several resources, including hardware components, software tools, cloud services, and development effort. The following section describes the estimated costs required to support the design, implementation, and evaluation of the proposed system.

7.1 Hardware Costs

Several hardware components are required to capture video data and support system processing.

Item	Estimated Cost (LKR)	Justification
Logitech C270 Webcam	10,000	Used to capture video streams for face detection and skeletal pose analysis during system testing.
Logitech C920 Webcam	25,000	Higher resolution camera to improve face recognition and pose estimation accuracy.
Generic 1080p USB Webcam	8,000	Additional camera used to test system compatibility with different camera devices.
Edge Processing Device / Development PC	0	Existing personal computer used for system development and testing.
External Storage Device	15,000	Used to store identity profiles, experimental data, and monitoring logs.

Table 1:Hardware Cost

Estimated Hardware Cost: 33,000 – 58,000 LKR

Justification:

The hardware components are required to capture video input and perform real-time computer vision processing. Multiple webcam devices are used to evaluate system performance under different camera conditions. The development computer already available reduces overall hardware expenses. Storage devices are required to store system logs, facial embeddings, and skeletal data used for testing and evaluation.

7.2 Software Licenses

Software	Cost (LKR)	Justification
Python	0	Open-source programming language used for system development.
OpenCV	0	Open-source computer vision library used for video processing and image analysis.
TensorFlow / PyTorch	0	Open-source machine learning frameworks used to implement deep learning models.
OpenPose / Pose Estimation Libraries	0	Used for skeletal pose detection from video frames.
VS Code / Development Tools	0	Open-source development environment used for coding and debugging.

Table 2:Software Licenses

Estimated Software Cost: 0 LKR

Justification:
The proposed system relies entirely on open-source technologies, which eliminates licensing costs. Python-based machine learning frameworks and computer vision libraries provide powerful tools for developing face recognition and skeletal pose estimation systems without requiring commercial software licenses.

7.3 Cloud Services

Assumptions:

- Basic cloud instance for remote monitoring services
- Approximate monthly cost: 8,000 LKR
- Usage duration: 6 months

Service	Monthly Cost (LKR)	Duration	Total Cost (LKR)
Cloud VM Hosting	8,000	6 months	48,000
Domain Registration	4,000	One-time	4,000

Notification Services (SMS / API)	2,000	6 months	12,000

Table 3:Cloud Service

Estimated Cloud Cost: 64,000 LKR

Justification:
The cloud infrastructure supports several system components including remote monitoring dashboards, notification services, system logs, and backup storage. The cloud server will host the application interface and allow family members or administrators to access monitoring results remotely. Notification services enable alert messages to be sent when abnormal activities are detected.

7.4 Human Effort Cost

Although this project is developed as an academic research project, the development effort can be estimated to represent the actual work required to implement the system.

Activity	Estimated Hours
Dataset collection and preparation	60 hrs
Face recognition model implementation	80 hrs
Pose estimation and skeletal feature extraction	80 hrs
System integration and development	90 hrs
System testing and evaluation	50 hrs
Total Estimated Effort	360 hours

Table 4:Human Effort Cost

Cost Estimation

Assumptions:

- Intern rate = 1,500 LKR per day
- 1 working day = 8 hours

Total working days:

- 360 hours ÷ 8 hours/day = 45 days
- Estimated Human Effort Cost:
- 45 days × 1,500 LKR/day = 67,500 LKR

7.5 Total Estimated Budget

Category	Estimated Cost (LKR)
Hardware	33,000 – 58,000
Software	0
Cloud Hosting & Deployment	64,000
Human Effort (Intern basis)	67,500
Total Estimated Direct Cost	164,500 – 189,500 LKR

Table 5:Total Estimated Budget

Overall Budget Justification

The proposed budget is designed to keep development costs relatively low while still providing the necessary resources for building and testing the elder-care monitoring system. Cost efficiency is achieved through the use of commonly available webcam hardware and open-source software frameworks for computer vision and machine learning development.

The main expenses arise from hardware devices used for video capture and cloud services used for remote monitoring and notification delivery. Human effort represents the time required for system design, algorithm implementation, system integration, and performance evaluation.

This approach demonstrates that the proposed system can be developed with moderate financial resources while still maintaining strong technical capabilities. The cost structure also indicates that the system could be economically viable for future deployment in elder-care facilities and home monitoring environments.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Overview of Work Breakdown

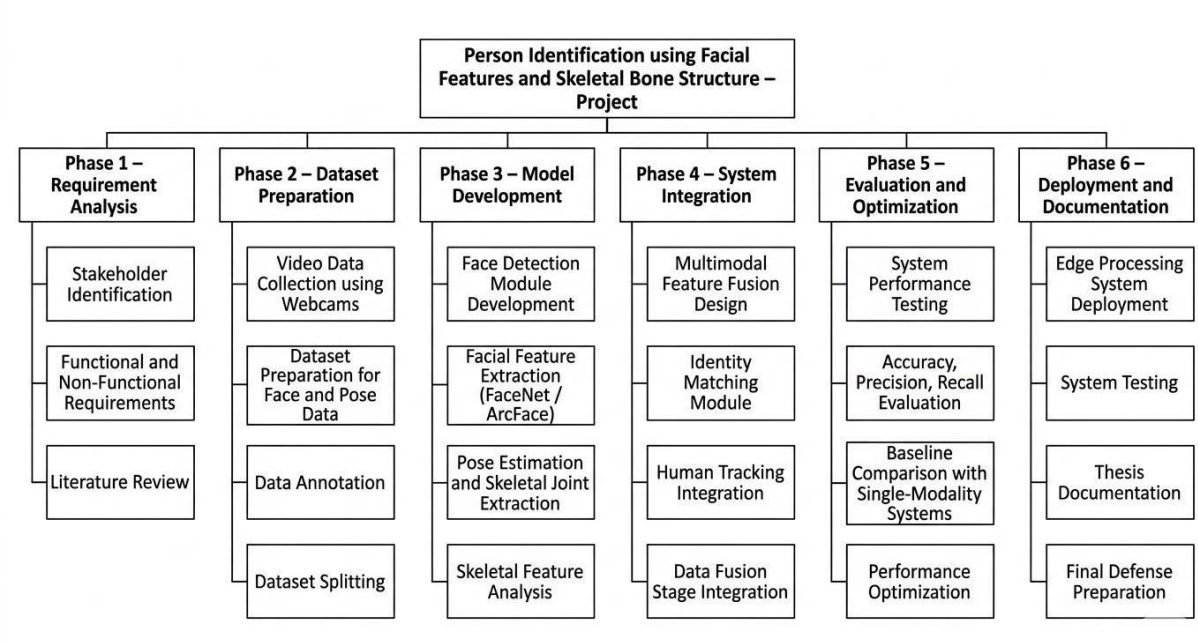


Figure 2:work breakdown structure

presents the hierarchical Work Breakdown Structure (WBS) of the proposed person identification system. The project is organized into six major phases that guide the development process from requirement analysis to final deployment and documentation. Each phase includes clearly defined tasks to ensure a structured and systematic implementation process. The development focuses on integrating facial recognition and skeletal pose analysis for reliable person identification in elder-care monitoring environments. The WBS also highlights the individual research component, including facial feature extraction, skeletal joint analysis, and multimodal feature fusion, which contribute to the core functionality of the system.

8.2 Timeline Alignment

The project spans twelve months following a phased development schedule:

- **Months 1–2:** Requirement analysis, stakeholder identification, and literature review
- **Months 2–3:** Dataset collection using webcam cameras and initial data preparation
- **Month 3:** Data annotation and dataset preparation for model training
- **Months 4–6:** Development of face recognition and skeletal pose estimation modules
- **Months 6–7:** Multimodal feature fusion and system integration
- **Months 7–9:** System testing, experimental evaluation, and performance optimization

- **Months 9–10:** System deployment and monitoring environment testing
- **Months 10–12:** Documentation, thesis writing, and final defense preparation

This timeline supports iterative development, testing, and system refinement while ensuring the project remains feasible within the academic schedule.

8.3 Task Breakdown Per Member

Phase	Member 1 – Person Identification (Face + Skeletal)	Member 2 – Human Tracking	Member 3 – Pose Estimation	Member 4 – Anomaly Detection & Monitoring
Requirement Analysis	Define objectives for facial recognition and skeletal feature-based identification	Define objectives for human movement tracking	Define pose estimation objectives and skeletal data extraction	Define anomaly detection and schedule monitoring objectives
Dataset Preparation	Prepare face image dataset and skeletal joint data for identification	Prepare tracking dataset with movement patterns	Prepare pose estimation dataset with body joint annotations	Prepare activity and anomaly detection dataset
Model Development	Implement face detection, facial feature extraction (FaceNet / ArcFace), and skeletal feature analysis	Implement human tracking algorithms to monitor movement and location	Implement pose estimation model to detect skeletal joints	Implement pose-based anomaly detection algorithms
System Integration	Integrate multimodal feature fusion and identity matching module	Integrate tracking module with system data flow	Integrate pose estimation module with skeletal data pipeline	Integrate anomaly detection and schedule monitoring modules
Evaluation	Evaluate identification accuracy using metrics such as precision, recall, and F1-score	Evaluate tracking accuracy and movement detection reliability	Evaluate skeletal joint detection accuracy	Evaluate anomaly detection performance and system alerts

Table c:Task Breakdown Per Member

8.4 Balanced workload distribution

Workload distribution is structured to ensure fairness and technical parity among team members. While the Sinhala Grammar Checker involves both neural fine-tuning and rule based development, other components require comparable effort in dataset preparation, model adaptation, evaluation, and integration. Shared tasks such as API implementation, routing logic, and system testing further balance technical contribution. This structured allocation ensures that each member contributes meaningfully to both individual research objectives and overall platform development.

9. GANTT CHART

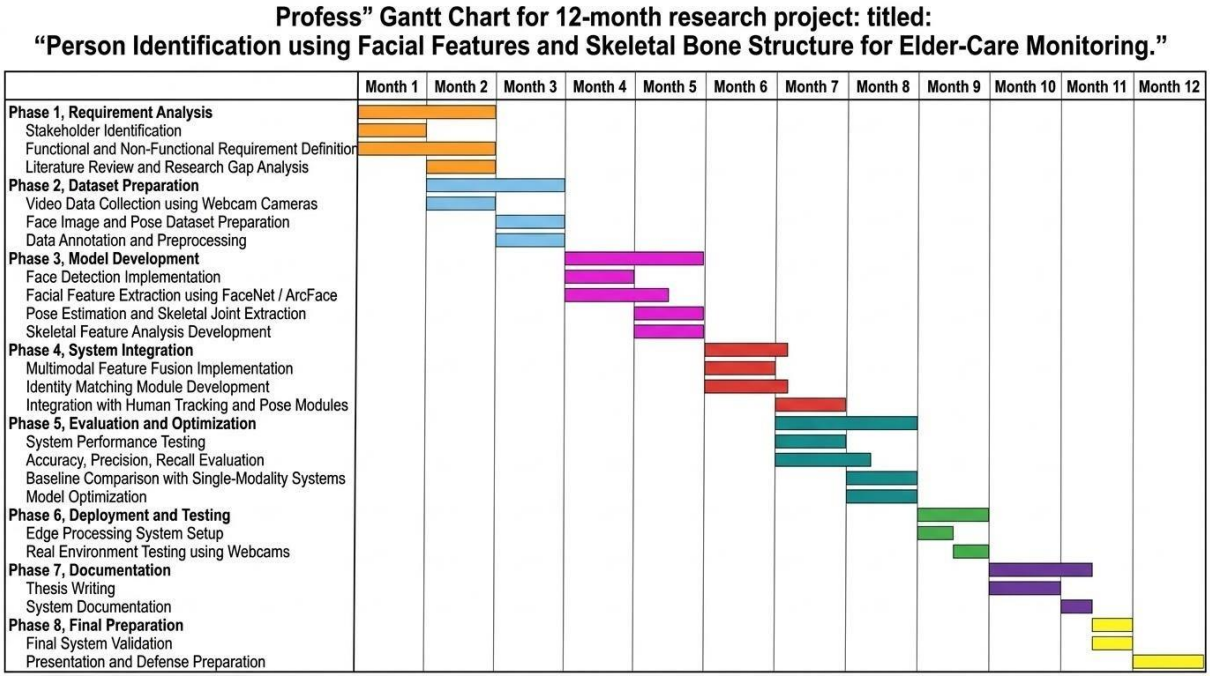


Figure 3:Gantt Chart

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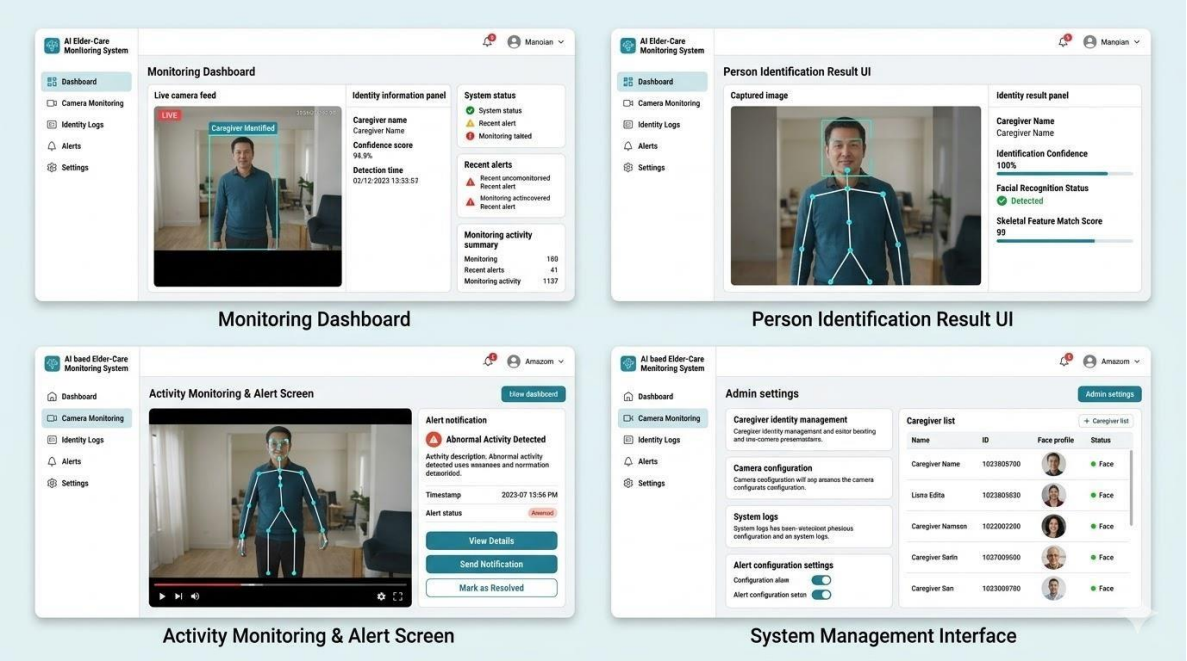
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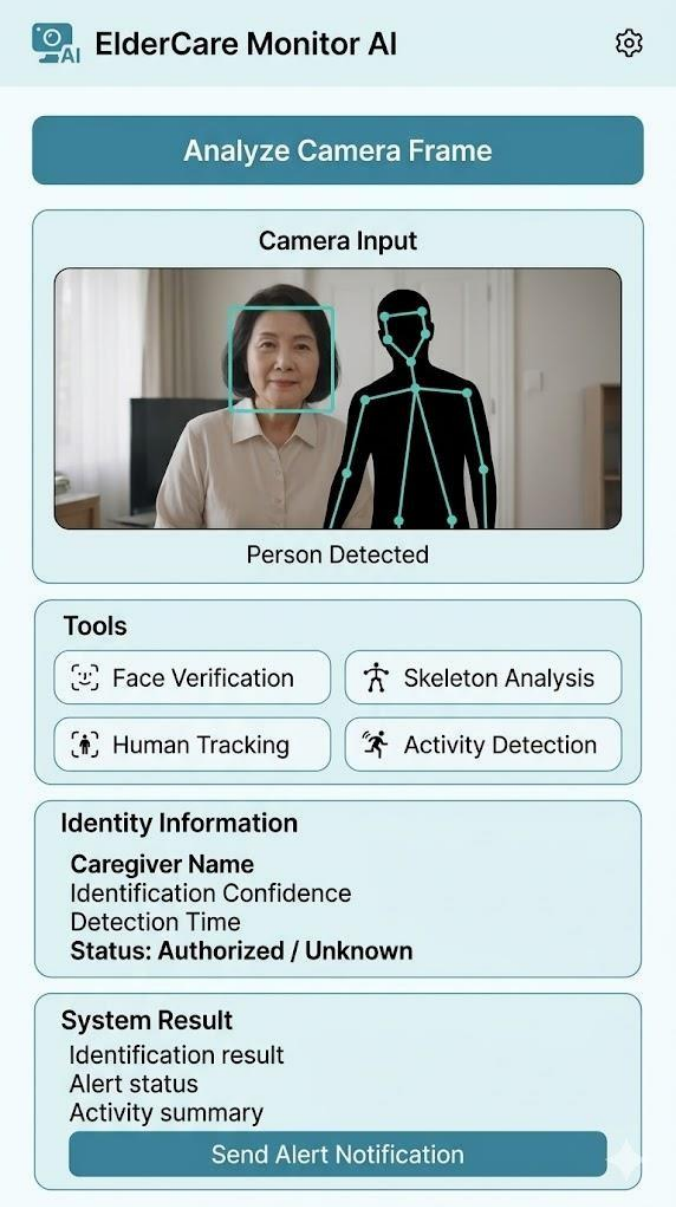
11. APPENDICES

11.1 Preliminary UI mockups

Appendix 1:Preliminary UI mockups



Web Extension UI



11.2 Risk Analysis

Appendix 2:Risk analysis

Appendix 2: Risk Analysis

The development of the **Person Identification using Facial Features and Skeletal Bone Structure system for elder-care monitoring** involves several potential risks that may affect system accuracy, reliability, and deployment. These risks are identified and mitigation strategies are proposed to ensure stable system performance.

1. Technical Risk

The person identification system may experience reduced accuracy in situations such as poor lighting conditions, occluded faces, or unusual camera angles. These factors may affect both facial recognition and skeletal pose estimation performance.

Mitigation:
To reduce this risk, the system will use a **multimodal identification approach** that combines facial features and skeletal body structure information. This allows the system to rely on skeletal features when facial information is unclear. Additional preprocessing techniques and model tuning will also be applied to improve robustness.

2. Pose Estimation Accuracy Risk

Pose estimation algorithms may sometimes produce incorrect skeletal joint detection, especially when multiple people appear in the camera frame or when body parts are partially hidden.

Mitigation:
Reliable pose estimation frameworks will be used and tested with different datasets to improve accuracy. Post-processing techniques and confidence thresholds will be applied to filter unreliable skeletal detections.

3. False Identification Risk

The system may incorrectly identify individuals or generate false positives when the facial recognition model produces similar embeddings for different individuals.

Mitigation:
Confidence thresholds will be applied to identity matching results, and multimodal feature fusion will be used to combine facial and skeletal features before making final identification decisions. Additional verification mechanisms may also be introduced.

4. Privacy and Ethical Risk

The system processes sensitive biometric data such as facial features and skeletal body structure information, which may raise privacy and ethical concerns.

Mitigation:
Strict access control mechanisms will be implemented to protect stored identity data. Data will only be used for monitoring purposes, and informed consent will be obtained from individuals involved in data collection.

5. Infrastructure and Performance Risk

Real-time processing of video streams and deep learning models may require significant computational resources. Limited hardware resources may affect system performance during real-time monitoring.

Mitigation:
The system will use **edge processing and optimized computer vision models** to reduce computational requirements. Lightweight models and efficient frameworks will be used to ensure that the system operates efficiently on available hardware.

Overall, the identified risks are manageable through careful system design, robust computer vision techniques, and responsible data handling practices. These mitigation strategies help ensure reliable system performance and safe deployment in elder-care monitoring environments.

11.3 Ethical documents

Appendix 3:Ethical documents

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Match Groups

68 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

9 Missing Quotations 1%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
3% Publications
7% Submitted works (Student Papers)